Ms. Welles **Alg 2 Trig/Honors**

Chp 7.3: Double-Angle, Half-Angle, and Product-Sum Formulas

The **Double-Angle Formulas** allow us to find the values of the trigonometric functions at $2x$ from their values at x . The **Half-Angle Formulas** relate the values of the trigonometric functions at $(1/2)x$ to their values at x. The **Product-Sum Formulas** relate products of sines and cosines to sums of sines and cosines.

I. Double-Angle Formulas

$$\tan(x+x) = \frac{\tan x + \tan x}{1 - \tan x \cdot \tan x} = \frac{2\tan x}{1-\tan^2 x}$$

$$\sin 2x = 2\sin x\cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$= 1 - 2\sin^2 x$$

$$= 2\cos^2 x - 1$$

$$\tan 2x = \frac{2\tan x}{1-\tan^2 x} \quad \Big\} \frac{\sin x}{\cos x}$$

II. Half-Angle Formulas

half the interval to determine the quadrant and sign of answer

$$\sin\frac{u}{2} = \pm\sqrt{\frac{1-\cos u}{2}} \qquad \cos\frac{u}{2} = \pm\sqrt{\frac{1+\cos u}{2}} \qquad \tan\frac{u}{2} = \frac{1-\cos u}{\sin u} = \frac{\sin u}{1+\cos u}$$

Formulas for Lowering Powers

The Formulas for Lowering Powers allow us to write any trigonometric expression involving even powers of sine and cosine in terms of the first power of cosine only.

remember for AB and BC

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \cos^2 x = \frac{1+\cos 2x}{2}, \qquad \tan^2 x = \frac{1-\cos 2x}{1+\cos 2x}$$

$\cos 2x = 1 - 2\sin^2 x$
$2\sin^2 x = 1 - \cos^2 x$
$\sin^2 x = \frac{1}{2}(1-\cos 2x)$

$\cos^2 x = 2\cos^2 x - 1$
$1 + \cos 2x = 2\cos^2 x$
$\frac{1}{2}(1+\cos 2x) = \cos^2 x$

III. Product-Sum Formulas

Product-to-Sum Formulas.

$$\sin u\cos v = \frac{1}{2}[\sin(u+v)+\sin(u-v)] \qquad \cos u\sin v = \frac{1}{2}[\sin(u+v)-\sin(u-v)]$$

$$\cos u\cos v = \frac{1}{2}[\cos(u+v)+\cos(u-v)] \qquad \sin u\sin v = \frac{1}{2}[\cos(u-v)-\cos(u+v)]$$

Sum-to-Product Formulas.

$$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2} \qquad \sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$$

$$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2} \qquad \cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$$

Example 1: Find $sin\ 2x$, $cos\ 2x$, and $tan\ 2x$ from the given information.

a. $tan\ x = \frac{-4}{3}$, $x\ in\ Quadrant\ II$

$\sin 2x = 2\sin x \cos x$

$= 2\left(\frac{4}{5}\right)\left(-\frac{3}{5}\right)$

$= \boxed{-\frac{24}{25}}$

$\cos 2x = \cos^2 x - \sin^2 x$

$= \left(-\frac{3}{5}\right)^2 - \left(\frac{4}{5}\right)^2$

$= \boxed{-\frac{7}{25}}$

$\tan 2x = \frac{\sin 2x}{\cos 2x}$

$= \frac{-\frac{24}{25}}{-\frac{7}{25}}$

$= \boxed{\frac{24}{7}}$

b. $csc\ x = 4,\ tan\ x < 0$

QII

$\frac{r}{y} = \frac{4}{1}$

$x = \sqrt{16-1}$

$= -\sqrt{15}$

$\sin 2x = 2\sin x \cos x$

$= 2\left(\frac{1}{4}\right)\left(-\frac{\sqrt{15}}{4}\right)$

$= \boxed{\frac{-\sqrt{15}}{8}}$

$\cos 2x = 1 - 2\sin^2 x$

$= 1 - 2\left(\frac{1}{4}\right)^2 = 1 - 2\left(\frac{1}{16}\right)$

$= 1 - \frac{1}{8}$

$= \boxed{\frac{7}{8}}$

$\tan 2x = \frac{\sin 2x}{\cos 2x} = \frac{-\frac{\sqrt{15}}{8}}{\frac{7}{8}}$

$= \boxed{\frac{-\sqrt{15}}{7}}$

Example 2:Use the formulas for lowering powers to rewrite the expression in terms of the first power of cosine.

$\cos^2 2x = \frac{1}{2}(1 + \cos 4x)$

$sin^4 x = \sin^2 x \cdot \sin^2 x$

$= \frac{1}{2}(1 - \cos 2x) \cdot \frac{1}{2}(1 - \cos 2x)$

$= \frac{1}{4}(1 - 2\cos 2x + \underbrace{\cos^2 2x}_{\frac{1}{2}(1+\cos 4x)})$

$= \boxed{\frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{8} + \frac{1}{8}\cos 4x}$

Example 3:Use an appropriate Half-Angle Formula to find the exact value of the expression.

a. $cos\ \frac{3\pi}{8}$

$\cos\frac{3\pi}{8} = \sqrt{\frac{1 + \cos\frac{3\pi}{4}}{2}}$

$= \boxed{\sqrt{\frac{1 - \frac{\sqrt{2}}{2}}{2}}}$

b. $tan\ 22.5^\circ$

$\frac{\theta}{2} = 22.5^\circ$

$\theta = 45^\circ$

$\tan 22.5^\circ = \frac{\sin 45^\circ}{1 + \cos 45^\circ}$

$= \boxed{\frac{\frac{\sqrt{2}}{2}}{1 + \frac{\sqrt{2}}{2}}}$

Example 4: Simplify the expression by using a Double-Angle Formula or a Half-Angle Formula.

a. $2\ sin\left(\frac{\theta}{2}\right) cos\left(\frac{\theta}{2}\right)$

$= \sin 2\left(\frac{\theta}{2}\right)$

$= \sin\theta$

b. $\frac{2\ tan\ 5\theta}{1-tan^2 5\theta}$

$\tan 2(5\theta)$

$\tan(10\theta)$

c. $\sqrt{\frac{1-cos\ 8\theta}{2}}$

$\sin 4\theta$, Q2, Q1

Example 5: Find $sin\ \frac{x}{2}$, $cos\ \frac{x}{2}$, and $tan\ \frac{x}{2}$ from the given information.

$\frac{x}{r} = \frac{-4}{5}$

$y = 3$

a. $cos\ x = \frac{-4}{5}$, $\frac{180^\circ}{2} < \frac{x}{2} < \frac{270^\circ}{2}$ → Q3

→ $90^\circ < \frac{x}{2} < 135^\circ$ → Q2

$\sin\frac{x}{2} - \sqrt{\frac{1-\cos x}{2}} = \sqrt{\frac{1+\frac{4}{5}}{2}} = \boxed{\frac{3\sqrt{10}}{10}}$

$\cos\frac{x}{2} = -\sqrt{\frac{1+\cos x}{2}} = -\sqrt{\frac{1-\frac{4}{5}}{2}} = \boxed{\frac{-\sqrt{10}}{10}}$

$\tan\frac{x}{2} = \frac{\sin\frac{x}{2}}{\cos\frac{x}{2}} = \frac{\frac{3\sqrt{10}}{10}}{\frac{-\sqrt{10}}{10}} = \boxed{-3}$

Example 6: Find the exact value of the given expression.

$\frac{y}{x} = \frac{12}{5}$

$r = 13$

$cos\left(2\ \underbrace{tan^{-1}\left(\frac{12}{5}\right)}_{\theta}\right)$ → $\cos 2\theta = 2\cos^2\theta - 1$

$= 2\left(\frac{5}{13}\right)^2 - 1$

$= 2\left(\frac{25}{169}\right) - 1$

$= \frac{50}{169} - 1$

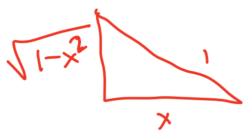
$= \boxed{-\frac{119}{169}}$

Example 7: Write the given expression as an algebraic expression in x.

$tan\left(2cos^{-1}x\right)$

$\theta = \cos^{-1} x$

$\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$



$$\tan 2\theta = \boxed{\dfrac{2\left(\frac{\sqrt{1-x^2}}{x}\right)}{1-\left(\frac{\sqrt{1-x^2}}{x}\right)^2}}$$

Example 8: Evaluate each expression under the given conditions.

a. $sin\left(\frac{\theta}{2}\right)$; $tan\ \theta = \frac{-5}{12}$, θ in Quadrant IV

$\frac{y}{x} = \frac{-5}{12}$

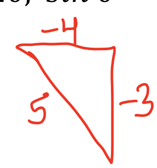
$r = 13$

$$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\left(\frac{12}{13}\right)}{2}}$$

$$= \sqrt{\frac{1}{26}}$$

$$= \boxed{\frac{\sqrt{26}}{26}}$$

b. $cos\ 2\theta$; $sin\ \theta = \frac{-3}{5}$, θ in Quadrant III



$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$= \left(\left(-\frac{4}{5}\right)^2 - \left(-\frac{3}{5}\right)^2\right)$$

$$= \boxed{\frac{7}{25}}$$

Example 9: Find the product or sum.

a. $sin\ 2x\ sin\ 3x$

$$= \frac{1}{2}\left[\cos(2x-3x) - \cos(2x+3x)\right]$$

$$= \frac{1}{2}\left[\cos(-x) - \cos(5x)\right]$$

$$= \boxed{\frac{1}{2}\left[\cos x - \cos(5x)\right]}$$

c. $3\ cos\ 37.5^{\circ}\ cos\ 7.5^{\circ}$

$$\frac{3}{2}\left[\cos(37.5^\circ + 7.5^\circ) + \cos(37.5^\circ - 7.5^\circ)\right]$$

$$= \frac{3}{2}\left[\cos(45^\circ) + \cos(30^\circ)\right]$$

$$= \boxed{\frac{3}{2}\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}\right)}$$

b. $cos\ \frac{\pi}{12} + cos\ \frac{5\pi}{12}$

$$= 2\cos\left(\frac{\frac{\pi}{12}+\frac{5\pi}{12}}{2}\right)\cos\left(\frac{\frac{\pi}{12}-\frac{5\pi}{12}}{2}\right)$$

$$= 2\cos\left(\frac{\pi}{4}\right)\cos\left(-\frac{\pi}{6}\right)$$

$$= 2\cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{6}\right)$$

$$= 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \boxed{\frac{\sqrt{6}}{2}}$$

d. $sin\ 2x - sin\ 7x$

$$= 2\left[\cos\frac{(2x+7x)}{2}\sin\frac{(2x-7x)}{2}\right]$$

$$= 2\cos\left(\frac{9}{2}x\right)\sin\left(-\frac{5}{2}x\right)$$

$$= \boxed{-2\cos\left(\frac{9}{2}x\right)\sin\left(\frac{5}{2}x\right)}$$

Example 10: Prove the Identity

a. $\sin 8x = 2\sin 4x \cos 4x$

RHS : $2\sin 4x \cos 4x$

$= \sin(2(4x))$

$= \sin 8x$

$= LHS$

b. $\cos^4 x - \sin^4 x = \cos 2x$

$(\cos^2 x - \sin^2 x)\underbrace{(\cos^2 x + \sin^2 x)}_{1}$

$= \cos^2 x - \sin^2 x$

$= \cos 2x$

$= RHS$

c. $\frac{1-\cos 2x}{\sin 2x} = \tan x$

LHS : $\frac{1-\cos 2x}{\sin 2x}$

$= \frac{1-(1-2\sin^2 x)}{\sin 2x}$

$= \frac{1-1+2\sin^2 x}{\sin 2x}$

$= \frac{2\sin^2 x}{\sin 2x}$

$= \frac{2\sin^2 x}{2\sin x \cos x}$

$= \frac{\sin x}{\cos x} = \tan x = RHS$

d. $\tan\frac{x}{2} + \cos x \tan\frac{x}{2} = \sin x$

LHS : $\frac{1-\cos x}{\sin x} + \cos x\left(\frac{1-\cos x}{\sin x}\right)$

$= \frac{1-\cos x + \cos x - \cos^2 x}{\sin x}$

$= \frac{1-\cos^2 x}{\sin x}$

$= \frac{\sin^2 x}{\sin x}$

$= \sin x$

$= RHS$

e. $\frac{\sin 3x + \sin 7x}{\cos 3x - \cos 7x} = \cot 2x$

$= \frac{2\left[\cos\left(\frac{-4x}{2}\right)\cancel{\sin\left(\frac{10x}{2}\right)}\right]}{-2\left[\cancel{\sin\left(\frac{10x}{2}\right)}\sin\left(\frac{-4x}{2}\right)\right]}$

$= -\frac{\cos(-2x)}{\sin(-2x)}$

$= \frac{-\cos 2x}{-\sin 2x}$

$= \cot 2x$

$= RHS$

f. $\tan^2\left(\frac{x}{2} + \frac{\pi}{4}\right) = \frac{1+\sin x}{1-\sin x}$

LHS : $\tan^2\left(\frac{x}{2} + \frac{\pi}{4}\right)$

$\tan^2\left(\frac{1}{2}\left(x + \frac{\pi}{2}\right)\right)$

$\frac{1-\cos\left(x+\frac{\pi}{2}\right)}{1+\cos\left(x+\frac{\pi}{2}\right)}$

$\frac{1+\sin(x)}{1-\sin(x)}$

$= RHS$